

MSTAR PRACTICE WORKSHEET**Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents**

Grade 6 Mathematics · Standard 6.AT.6c

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: BEvaluate $3 + 4 \times 2$.

- ☐ A 9
- ☒ B 11
- ☐ C 14
- ☐ D 24

Multiplication comes before addition: $4 \times 2 = 8$, then $3 + 8 = 11$.

Question 2 SELECTED RESPONSE — CORRECT: AWhat does 2^3 mean?

- ☒ A $2 \times 2 \times 2$
- ☐ B 2×3
- ☐ C $2 + 2 + 2$
- ☐ D 3×3

The exponent says how many times the base is used as a factor, so $2^3 = 2 \times 2 \times 2 = 8$.

Question 3 **SELECTED RESPONSE — CORRECT: A**

Sofia buys 4 tickets at \$20 each and 2 tickets at \$15 each. Which expression shows the total?

- ☒ A $4 \times 20 + 2 \times 15$
- ☐ B $4 + 20 + 2 + 15$
- ☐ C $(4 + 2) \times (20 + 15)$
- ☐ D $4 \times 2 + 20 \times 15$

Each ticket type is a count times a price, and the two costs are added: $4 \times 20 + 2 \times 15$.

Question 4 **SELECTED RESPONSE — CORRECT: C**

Evaluate $5^2 + 3$.

- ☐ A 13
- ☐ B 16
- ☒ C 28
- ☐ D 64

Powers come first: $5^2 = 25$. Then add: $25 + 3 = 28$.

Question 5 **SELECTED RESPONSE — CORRECT: B**

What is the total of $2 \times 15 + 4 \times 20 + 8 \times 2^3$?

- ☐ A 126
- ☒ B 174
- ☐ C 210
- ☐ D 1,392

Power first: $2^3 = 8$. Multiply: 30, 80, 64. Add: $30 + 80 + 64 = 174$.

Question 6

SELECTED RESPONSE — CORRECT: D

Miguel evaluates $12 - 2 \times 3$ and gets 30. Margaret gets 6. Who is right?

- ☐ A Miguel, because he worked left to right
- ☐ B Both are right
- ☐ C Neither is right
- ☒ D Margaret, because multiplication comes before subtraction

Multiplying first gives $2 \times 3 = 6$, then $12 - 6 = 6$. Miguel subtracted first, getting $10 \times 3 = 30$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

Evaluate the numerical expression $3 \times 4^2 - 8$.

- ☐ A 16
- ☐ B 24
- ☒ C 40
- ☐ D 136

Following the order of operations, evaluate the power first: $4^2 = 16$. Then multiply: $3 \times 16 = 48$. Then subtract: $48 - 8 = 40$.

- **A:** Only the power was evaluated. Continue: multiply by 3, then subtract 8.
- **B:** The exponent was treated as a multiplier, 4 times 2. Four squared is 16.
- **D:** The multiplication happened before the power. Powers come first: evaluate 4 squared, then multiply.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- ☐ A Multiply 3×4 first, then square the product: $12^2 = 144$, and $144 - 8 = 136$.
- ☒ B Evaluate the power first: $4^2 = 16$. Then multiply: $3 \times 16 = 48$. Then subtract: $48 - 8 = 40$.
- ☐ C Read 4^2 as $4 \times 2 = 8$, then multiply $3 \times 8 = 24$ and subtract: $24 - 8 = 16$.
- ☐ D Subtract first because 16 and 8 are next to each other: $16 - 8 = 8$, then $3 \times 8 = 24$.

The order of operations puts exponents before multiplication, and multiplication before subtraction. Each other path breaks that order.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Subtracting Too Soon

Devon evaluated $20 - 2 \times 3^2$. His work was: $3^2 = 9$, then $20 - 2 = 18$, then $18 \times 9 = 162$. He wrote that the value of the expression is 162.

Explain Devon's misconception and give the correct value of the expression.

Model answer: Devon evaluated the power correctly but then subtracted before multiplying. In PEMDAS, multiplication comes before subtraction: $3^2 = 9$, then $2 \times 9 = 18$, then $20 - 18 = 2$. The value is 2.

2 points	Student explains that the order of operations requires multiplying 2×9 before subtracting from 20, and gives the value 2.
1 point	Student gives the value 2 but does not explain that multiplication comes before subtraction.
0 points	Student agrees with Devon that the value is 162 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Seven friends split the bill for a pizza party. They order 4 pizzas at 3^2 dollars each, plus a one-time \$6 delivery fee. Write ONE numerical expression for what each friend owes, evaluate it, and explain why the grouping symbols matter.

Model answer: Each friend owes $(4 \times 3^2 + 6) \div 7$. Evaluating: the power first, $3^2 = 9$; then multiply, $4 \times 9 = 36$; then add inside the grouping symbols, $36 + 6 = 42$; then divide, $42 \div 7 = \$6$ each. The grouping symbols matter because without them the expression would read $4 \times 3^2 + 6 \div 7$, which divides only the delivery fee among the friends and leaves each of them paying for all four pizzas — about \$36.86 instead of \$6. The parentheses are what say "split the WHOLE bill."

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.